

Anil Sahith Vallepu

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PROFESSIONAL SUMMARY

AI/ML Engineer with strong Math, Software Engineering fundamentals, and experience in building intelligent systems that bridge research with practical applications, along with a proven ability to optimise model performance for resource-constrained environments. Passionate about leveraging emerging technologies to create scalable, impactful solutions for real-world problems in healthcare, finance, automation, and enterprise workflows.

EDUCATION

- National Institute of Technology (NIT) Warangal** Warangal, India
Bachelor of Technology in Electrical and Electronics Engineering. May 2024

SKILLS

- Programming Languages:** Python, C++, JavaScript, TypeScript, SQL, MATLAB.
- Tools & Platforms:** VS Code, PyCharm, Figma, Cursor, Jupyter Notebook, Postman, GitHub, Vercel.
- Web Technologies:** React.js, Next.js, Node.js, Flask, FastAPI, RESTful APIs, WebSocket, Streamlit, Gradio.
- Databases & Cloud:** MySQL, PostgreSQL, MongoDB, Redis, Pinecone, FAISS, ChromaDB, AWS, GCP, Docker.
- Machine Learning & AI:** TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, OpenCV, NLP, RAG, Knowledge Graphs, LLMs, Hugging Face Transformers, Langchain.
- Specialisations:** Data Structures and Algorithms, System Design, Embedded Systems, Edge Computing, LLM Fine-Tuning, Inference Optimisation, Quantum Computing, PQC, Quant Finance.

EXPERIENCE

- Noccarc Robotics Pvt Ltd** June 2024 - April 2025
AI Engineer Pune, Maharashtra
 - Architected a real-time arrhythmia detection system for embedded patient monitoring devices using deep learning, achieving 95%+ accuracy across 18+ cardiac arrhythmia classes on the PhysioNet database.
 - Implemented a hybrid neural network combining ResNet, CBAM attention mechanism, and GRU layers with separable convolutions for efficient edge deployment, optimised inference pipeline to process ECG data every 5 seconds while consuming under 12% single-core CPU resources.
 - Established an end-to-end data pipeline from ECG sensor integration to frontend visualisation and backend processing with automating alarm generation, created 20+ distinct alarm classifications for critical heart abnormalities, reducing clinical response times by 40% in critical care environments.
- Carelon Global Solutions (Elevance Healthcare)** June 2023 - August 2023
Summer Intern - AI/ML Bangalore, Karnataka
 - Automated document processing pipeline using computer vision and deep learning to extract and classify data from 700+ documents per hour with 95% accuracy; deployed an end-to-end OCR solution using PyTesseract for text extraction, OpenCV for image preprocessing, and YOLOv8 object detection for region identification.
 - Accelerated processing performance through multi-threading and multi-processing techniques, reducing document processing time by 90% and decreasing operational costs, delivered a production-ready solution by integrating with existing data workflows for scalable enterprise document management.

PROJECTS

- MiniAI-As-A-Service App:** Designed a full-stack SaaS application for automated CSV data analysis with machine learning model training and AI-generated insights; implemented FastAPI backend for model training, prediction, and statistical analysis endpoints with a Redis caching layer and PostgreSQL for persistent storage. [🔗 Link](#)
- SlackBot Content Creation AI Assistant:** Launched an AI-powered Slack bot automating content research, keyword analysis, and competitive analysis workflows; incorporated multiple APIs for web scraping, search engine results, and email notifications via SendGrid, reducing content planning time from 6+ hours to 15 minutes for marketing teams. [🔗 Link](#)
- HSN Code Classification System:** Constructed an intelligent chatbot system using RAG and Knowledge Graphs for accurate HSN code classification; leveraged NetworkX to model hierarchical HSN code relationships and deployed a user-friendly web interface using Streamlit with 92% classification accuracy. [🔗 Link](#)
- Dynamic AI Exam-Prep App:** Developed an adaptive learning platform with a multi-agent orchestration system for personalised NCLEX exam preparation; programmed an intelligent scheduling algorithm for dynamic study calendar generation based on performance metrics and integrated a contextual AI chatbot tutor. [🔗 Link](#)

ACHIEVEMENTS

- Secured 99.3rd percentile in JEE Mains (India) - Top 0.7% among 1 million+ candidates
- Recognised with Top 100 Projects at NASA AMES Space Settlement Contest, San Juan, Puerto Rico (2017)
- Solved 300+ Data Structures and Algorithms problems (Medium-Hard difficulty) on LeetCode